FIITJEE

ALL INDIA TEST SERIES

PART TEST – III

JEE (Main)-2023

TEST DATE: 17-12-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical based questions. The answer to each question is rounded off to the nearest integer value. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

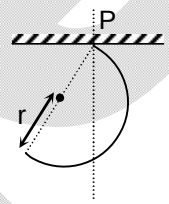
SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

1. The wave length of light incident on a metal surface is reduced from 300 nm to 200 nm (both are less than threshold wavelength). What is the change in the stopping potential for photoelectrons emitted from the surface (take h = 6.6×10^{-34} J-sec)
(A) 0.0625 V (B) 1.0625 V
(C) 2.0625 V (D) 3.0625 V

2. A uniform semicircular ring having mass m and radius r is hanging at one of its end (P) freely as shown in the given figure. The ring is slightly disturbed so that it oscillates in its own plane. The time period of oscillation of the ring is



(A) $2\pi\sqrt{\dfrac{r}{g\left(1+\dfrac{1}{\pi^2}\right)}}$ (B) $2\pi\sqrt{\dfrac{r}{g\left(1-\dfrac{4}{\pi^2}\right)^{1/2}}}$

(C) $2\pi\sqrt{\dfrac{r}{g\left(1-\dfrac{2}{\pi^2}\right)^{1/2}}}$ (D) $2\pi\sqrt{\dfrac{2r}{g\left(1+\dfrac{4}{\pi^2}\right)^{1/2}}}$

3. A number of little droplets of a liquid of surface tension T, density ρ, all of same radius r, combine to form a single drop of radius R. If the total energy released in this process is converted into kinetic energy of the bigger drop, velocity of the bigger drop is
(A) $\sqrt{\dfrac{3T(R-r)}{Rr\rho}}$ (B) $\sqrt{\dfrac{6T(R-r)}{Rr\rho}}$
(C) $\sqrt{\dfrac{T(R-r)}{Rr}}$ (D) $\sqrt{\dfrac{2T(R-r)\rho}{R}}$

4. In a screw gauge, 5 complete rotations of the screw cause it to move a linear distance of 0.25 cm and each rotation corresponds to one main scale division. There are 100 circular scale divisions. The thickness of a wire measured by this screw gauge gives a reading of 4 main scale divisions and 30 circular scale divisions. Assuming negligible zero error, what is the thickness (in mm) of the wire?
(A) 2.15 (B) 3.15
(C) 1.25 (D) 2.25

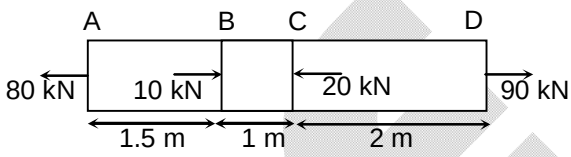
5. The refracting angle of a prism is A and the refractive index of the material of the prism is $\cot\left(\dfrac{A}{2}\right)$. The angle of minimum deviation is
(A) 180° – 3A (B) 180° + 2A
(C) 90° – A (D) 180° – 2A

6. In a hypothetical system, a particle of mass m and charge $-3q$ is moving around a very heavy particle having charge q. Assume that Bohr's model is applicable to this system, then velocity of mass m in the first orbit is

(A) $\frac{3q^2}{2\varepsilon_0 h}$ (B) $\frac{3q^2}{4\varepsilon_0 h}$

(C) $\frac{3q}{2\pi\varepsilon_0 h}$ (D) $\frac{3q}{4\pi\varepsilon_0 h}$

7. A steel rod of cross sectional area 1 m^2 is acted upon by forces shown in figure. The elongation of the length BC of the rod is (Y = 2.0×10^{11} N/m^2)



(A) 13×10^{-7} m (B) 4.5×10^{-7} m

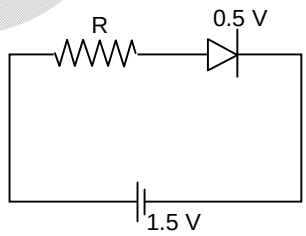
(C) 5×10^{-7} m (D) 3.5×10^{-7} m

8. A body executes SHM of period 20 seconds. Its velocity is 5 cm/s, 2 seconds after it has passed the mean position. Find the amplitude (in cm) of the bob. (cos 36° = 0.809 and $\pi = \frac{22}{7}$)

(A) 15.67 (B) 17.67

(C) 19.67 (D) 21.67

9. The diode used in the circuit shown in figure has a constant voltage drop of 0.5 V at all currents and a maximum power rating of 100 mW. What should be the value of the resistance R, connected in series with diode for obtaining maximum current?



(A) 5Ω (B) 5.6Ω

(C) 6.7Ω (D) 20Ω

10. A ray enters a glass slab at an angle θ from air and the refractive index varies along the slab as $\mu = \mu_0 - kt^2$, where t is the distance measured along the normal. μ_0 and k are positive constants. If the glass slab is sufficiently thick, how far along the normal will the ray go before it reflects back?

(A) $\frac{\mu_0 - \sin\theta}{k}$ (B) $\frac{\mu_0 + \sin\theta}{k}$

(C) $\sqrt{\frac{\mu_0 - \sin\theta}{k}}$ (D) $\sqrt{\frac{\mu_0 + \sin\theta}{k}}$

11. The electric field part of an electromagnetic wave in a medium is represented by

$E_x = 0$

$E_y = 2.5 \text{ N/C} \cos\left[\left(2\pi \times 10^6 \text{ rad/s}\right)t - \left(\pi \times 10^{-2} \text{ rad/m}\right)x\right]$

and $E_z = 0$

Then the wave is

(A) moving along x direction with frequency 10^6 Hz and wavelength 100 m

(B) moving along x direction with frequency 10^6 Hz and wavelength 200 m

(C) moving along $-x$ direction with frequency 10^6 Hz and wavelength 200 m

(D) moving along y direction with a frequency $2\pi \times 10^6$ Hz and wavelength 200 m.

12. In a Fraunhofer diffraction experiment at a single slit using a light of wavelength 400 nm, the first minimum is formed at an angle 30°. The direction θ of the first secondary maximum is given by:

(A) $\sin^{-1}\left(\frac{2}{3}\right)$ (B) $\sin^{-1}\left(\frac{3}{4}\right)$

(C) $\sin^{-1}\left(\frac{1}{4}\right)$ (D) $\sin^{-1}\left(\frac{2}{5}\right)$

13. A cylindrical vessel filled with water upto a height h weighs w and is resting on a horizontal plane. The side of wall of the vessel has a plugged circular hole touching the bottom. The coefficient of friction between the bottom of the vessel and the plane is μ. The minimum diameter of the hole so that the vessel just begins to move on the floor after the removal of the plug is:

(A) $\sqrt{\frac{\pi\mu w}{\rho hg}}$ (B) $\sqrt{\frac{\mu w}{2\pi\rho hg}}$

(C) $\sqrt{\frac{\mu w}{\pi\rho hg}}$ (D) $\sqrt{\frac{2\mu w}{\pi\rho hg}}$

14. The transverse displacement y(x, t) of a wave on a string is given by $y(x, t) = e^{-(ax^2+bt^2+2\sqrt{ab}xt)}$. This represents a …….

(A) wave moving along +x-axis with speed $\sqrt{\frac{a}{b}}$

(B) wave moving along –x-axis with speed $\sqrt{\frac{b}{a}}$

(C) standing wave of frequency $\sqrt{b}$

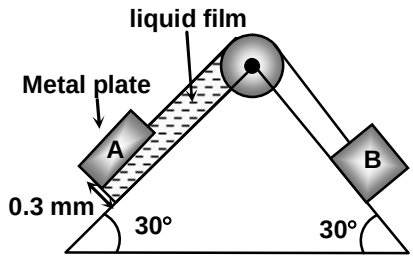
(D) standing wave of frequency $\frac{1}{\sqrt{b}}$

15. A body is fired with a velocity of magnitude $\sqrt{gR} < v < \sqrt{2gR}$ at an angle of 30° with the radius vector of the earth. If at the highest point, the speed of the body is $\frac{v}{4}$, the maximum height attained by the body is equal to (R = radius of earth and use the appropriate assumptions)

(A) $\frac{v^2}{8g}$ (B) R

(C) $R\sqrt{2}$ (D) 2R

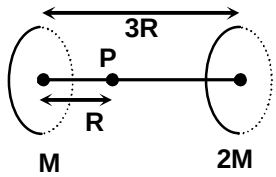
16. A metal plate of area 0.10 m^2 is connected to a 0.04 kg mass via a string that passes over an ideal pulley (considered massless and frictionless). A liquid with a film of thickness of 0.3 mm is placed between the plate and the table. When the system is released the plate moves up with constant speed of 0.085 m/s. Find the approximate value of coefficient of viscosity of the liquid. (mass of metal plate is 0.02 kg, g = 9.8 m/s^2)



(A) 3.46×10^{-3} Pa sec (B) 3.46×10^{-4} Pa sec

(C) 6.90×10^{-3} Pa sec (D) 6.90×10^{-4} Pa sec

17. Two rings having masses M and 2M, respectively, having the same radius are placed coaxially as shown in the figure. If the mass distribution on both the rings is non-uniform, then the gravitational potential at point P is



(A) $-\frac{GM}{R}\left[\frac{1}{\sqrt{2}}+\frac{2}{\sqrt{5}}\right]$

(B) $-\frac{GM}{R}\left[1+\frac{1}{\sqrt{2}}\right]$

(C) zero

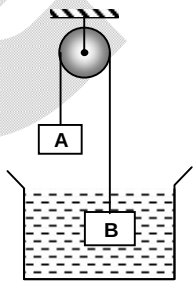
(D) cannot be determined from the given information

18. A heavenly body is receding from earth such that the fractional change in λ is 1, then its velocity is

(A) c (B) $\frac{3c}{5}$

(C) $\frac{c}{5}$ (D) $\frac{2c}{5}$

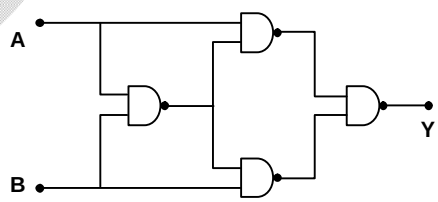
19. In the given figure $m_B = 3m$, density of liquid is ρ and density of block B is 2ρ. The system is released from rest so that block B moves up when in liquid and moves down when out of liquid with the same acceleration. Find the mass of block A.



(A) $\frac{7}{4}m$ (B) 2m

(C) $\frac{9}{7}m$ (D) $\frac{9}{4}m$

20. Truth table for a system of four NAND gates as shown is



(A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

(B)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	1

(C)

A	B	Y
0	0	1
0	1	1
1	0	0
1	1	0

(D)

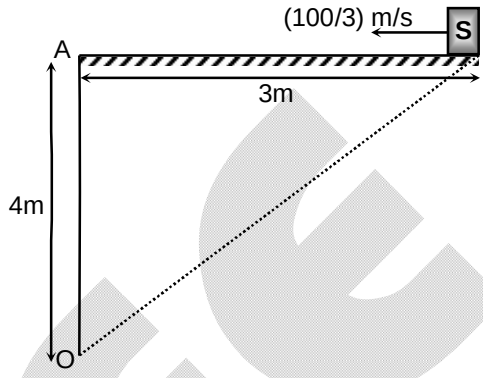
A	B	Y
0	0	1
0	1	0
1	1	0
1	1	1

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

21. A source of sound S is travelling at $\frac{100}{3}$ m/s along a road towards a point A. A person standing at point O on a road perpendicular to AS hears a sound of frequency f′ corresponding to the sound emitted at the instant when the source is 3m away from A. The distance of O from A is 4m. If the original frequency is 640 Hz, then find the value of f′ in Hz. (velocity of sound is 340 m/s)



22. A metallic sphere of radius 21 cm is kept in the path of a parallel beam of light of large aperture of intensity $\frac{1}{110}$ W/m^2. The force exerted by beam on the sphere is $k \times 10^{-13}$N. Find k. ($\pi = \frac{22}{7}$ and light is completely absorbed by sphere)

23. An astronomical telescope has objective and eyepiece of focal length 40 cm and 4 cm respectively. To view an object 200 cm away from the objective, at what distance the lenses must be separated (in cm)?

24. Cobalt-57 is radioactive, emitting β-particles. The half life for this is 270 days. If 100 mg of this is kept in an open container, then what mass (in mg) of Cobalt-57 will remain after 540 days.

25. A clear sheet of polaroid is placed on the top of similar sheet so that their axes make an angle $\sin^{-1}\left(\frac{3}{5}\right)$ with each other. The ratio of intensity of the emergent light to that of unpolarised incident light is $\frac{k}{25}$. Find k.

26. A narrow parallel beam of light is incident on a transparent sphere of refractive index $\frac{4}{n}$. If the beam finally gets focused at a point situated at a distance = 2 $\times$ (radius of sphere) from the centre of the sphere, then find n.

27. A hydrogen like atom of atomic number Z is in excited state of quantum number 2n. It can emit a maximum energy photon of 204 eV. If it makes a transition to the quantum state n, a photon of energy 40.8 eV is emitted. Calculate the atomic number Z. Ground state energy of hydrogen atom is –13.6 eV.

28. Image of an object approaching convex mirror of radius of curvature 20 m along its optical axis is observed to move from $\frac{25}{3}$ m to $\frac{50}{7}$ m in 30 sec. If the object is moving uniformly, then it's speed is n km/h. Find the value of n.

29. An open pipe is suddenly closed at one end with the result that the frequency of third harmonic of the closed pipe is found to be higher by 100 Hz than the fundamental frequency of the open pipe. Find the fundamental frequency of the open pipe in Hz.

30. In a double slit arrangement fringes are produced using light of wavelength 4800 Å. One slit is covered by a thin plate of glass of refractive index 1.4 and the other with another glass plate of same thickness but of refractive index 1.7. By doing so the central bright shifts to original fifth bright fringe from centre. Thickness of glass plate is k μm. Find the value of k.

FIITJEE

Chemistry

PART – B

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

31. The products obtained when I_2 reacts with hot and concentrated aqueous NaOH are
(A) IO^- and IO_3^-
(B) IO_2^- and IO_3^-
(C) I^- and IO^-
(D) I^- and IO_3^-

32. Which of the following product is not produced during the hydrolysis of XeF_4 ?
(A) XeO_3
(B) O_2
(C) XeO_2F_2
(D) Xe

33. Four successive members of first row transition element are listed below. Which one of them unexpectedly have highest $E^o_{M^{3+}/M^{2+}}$ value?
(A) Mn (Z = 25)
(B) Fe (Z = 26)
(C) V (Z = 23)
(D) Cr (Z = 24)

34. In context of actinoids, which of the following statements is not correct?
(A) The actinoids are radioactive elements which make their study more difficult.
(B) There is a gradual decrease in the radii of the members with increasing atomic number in the series.
(C) The magnetic properties of actinoids can not be explained easily, as they are more complex.
(D) As compare to lanthanoid compounds, actinoid compounds are less basic.

35. Which of the following has a square planar geometry?
(A) $[FeI_4]^{2-}$
(B) $[PdCl_4]^{2-}$
(C) $[NiCl_4]^{2-}$
(D) $[CoI_4]^{2-}$

36. Which of the following statements is correct?
(A) NO_2 is a reddish-orange gas and it does not play any role in photochemical smog.
(B) SO_2 does not affect the formation of chlorophyll in plants.
(C) CO combines with haemoglobin to form carboxyhaemoglobin which is more stable than oxyhaemoglobin.
(D) SO_2 is more harmful than SO_3.

37. The reaction that is involved in the ozone layer depletion mechanism in the stratosphere is

(A) $NO + \frac{1}{2}O_2 \xrightarrow{h\nu} NO_2$

(B) $CH_4 + O_2 \longrightarrow CH_2O + H_2O$

(C) $Cl^{\bullet} + CH_4 \longrightarrow \dot{C}H_3 + HCl$

(D) $ClO^{\bullet} + O \longrightarrow Cl^{\bullet} + O_2$

38. Which of the following chromates is not yellow in colour?

(A) $PbCrO_4$

(B) $SrCrO_4$

(C) Ag_2CrO_4

(D) $BaCrO_4$

39. Which of the following statements is correct?

(A) Dettol is a mixture of chloroxylenol and bithionol.

(B) 2-3% solution of chloroform in alcohol-water is known as tincture of iodine.

(C) Boric acid in concentrated aqueous solution is weak antiseptic for eyes.

(D) 0.2% solution of phenol is an antiseptic while 1% solution of phenol is disinfectant.

40. Which of the following antibiotic is baceteriostatic?

(A) Penicillin

(B) Tetracyclin

(C) Aminoglycosides

(D) Ofloxacin

41. Multimolecular colloid is present in

(A) Sol of starch

(B) Soap solution

(C) Sol of sulphur

(D) Sol of protein

42. 100 ml of 0.6 M acetic acid is shaken with 4 g activated carbon. The final concentration of the solution after adsorption is 0.4 M. The amount of acetic acid (in g) adsorbed per gram of carbon is……..?

(A) 0.6

(B) 0.3

(C) 1.2

(D) 0.9

43. Which of the following statements is incorrect?

(A) Average translational kinetic energy of a molecule in a gas at given temperature does not depend on its molecular mass.

(B) Root mean square speed of a molecule in a gas at a give temperature is halved when its molar mass is increased four times.

(C) Average translational kinetic energy of a molecule in a gas at a given temperature is doubled when its molar mass is increased twice.

(D) Root mean square speed of a molecule in a gas is doubled when its temperature is increased four times.

44. When 100 ml of 1.0 M HCl was mixed with 100 ml of 1.0 M NaOH in an insulated beaker at constant pressure, a temperature increase of $5.8^{o}C$ was measured for the beaker and its constants. If the enthalpy of neutralization of a strong acid with a strong base is -58.0 kJ mol^{-1}, the calorimeter constant in $\left(kJ\ k^{-1}\right)$ is……

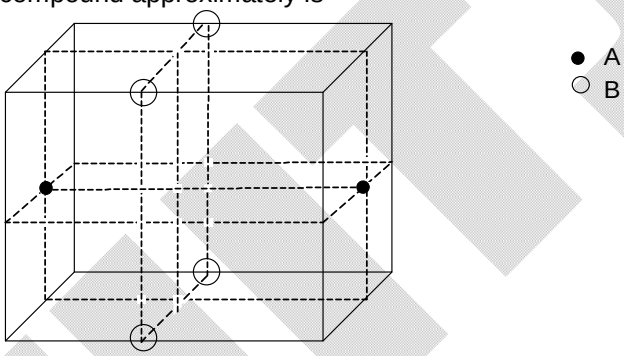
(Given: Heat capacity of solution is $4.2\ J\ g^{-1}k^{-1}$ and density of solution is $1\ g\ mL^{-1}$)

(A) 0.84
(B) 1
(C) 0.42
(D) 0.16

45. Which of the following ores has carbonate anion?
(A) Diaspore
(B) Calamine
(C) Willemite
(D) Argentite

46. Extraction of aluminium from bauxite does not involve which of the following steps:
(A) Crushing followed by concentration of the ore is done by Leaching process.
(B) Extraction of aluminium from concentrated ore is done by Hall Heroult process.
(C) Extraction of aluminium from concentrated ore is done by carbon.
(D) In reduction process CaF_2 is mixed with concentrated ore to reduce the melting point of the mixture and brings conductivity.

47. The cubic unit cell structure of compound containing cation A and anion B is shown below. The size of cation is smaller than anion and anion-anions are in contact. The packing efficiency of compound approximately is



(A) 73
(B) 55
(C) 85
(D) 25

48. The increase in the molar conductivity of HCl with dilution is due to
(A) Increase in the self ionization of water.
(B) Hydrolysis of HCl with dilution.
(C) Decrease in the self ionization of water.
(D) Decrease in the interionic forces by dilution.

49. The moles of $K_2Cr_2O_7$ are required to react completely with 2.4 mol of CuS in hot acidic medium
(A) 1.8
(B) 3.2
(C) 1.6
(D) 6.4

50. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is
(A) 2
(B) 5
(C) 4
(D) 3

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

51. The reaction of white phosphorous and excess of dry chlorine gas gives a compound Q. The number of moles of HCl produced by the complete hydrolysis of 1 mole of Q is_____

52. An aqueous solution is prepared by dissolving 0.2 mole of an ionic salt in 1.86 kg of water at 27^oC. The salt is only 80% dissociated in the solution. The freezing point of the solution is -0.52^oC. The number of ions present per formula unit of ionic salt is_____(Given molal depression constant of water (K_f) is 1.86 K kg mol^{-1} and freezing point of water is 0^oC).

53. The electrode potential of the following cell is – x at 300 K.
$Pt(s)\,|\,H_2(g)(4\ atm)\,/\,CH_3COOH(aq)(10^{-1}\ M)\,||\,NaOH(aq)(10^{-3}\ M)\,/\,H_2(g)(1\ atm)\,/\,Pt(s)$.
The value 1000x is_____
[Given: $\frac{2.303\ RT}{F}$ at 300 K = 0.06, log2 = 0.3, K_a for acetic acid is 10^{-5}]

54. A crystalline solid of a pure substance X has body centred cubic structure with a cell edge of 800 pm. The number of atoms present in 512 g of crystal X is $N\times10^{22}$ if the density of the substance in the crystal is 8 g cm^{-3}. The value of N is_____

55. Reaction of aqueous solution of KI with ozone produced a paramagnetic product Y. The spin only magnetic moment (in B.M.) of Y in nearest integer is_____

56. 60 ml of x M $AgNO_3$ required for complete precipitation of Br^- ions present in 15 ml of 0.02 M solution of $\left[Co(NH_3)_5Br\right]Br_2$ as silver bromide. The value 100x is_____

57. The enthalpy change (ΔH) for the following reaction at temperature 25^oC is -2573 kJ mol^{-1}.
$2C_2H_2(g)+5O_2(g)\longrightarrow4CO_2(g)+2H_2O(g)$
The average bond enthalpy of C – H bond is x kJ mol^{-1}, if the average bond enthalpy of $C\equiv C$, $O=O$, $C=O$ and O – H bonds are 814, 499, 724 and 640 kJ mol^{-1} respectively at temperature 25^oC. The x is….

58. On heating acidified NaCl with orange solid gives orange-red vapours. On passing these orange red vapours through NaOH solution turns it into yellow solution. This yellow solution gives yellow precipitate when the solution of acetic acid and lead acetate is added to this solution. The oxidation state of the element which has highest oxidation state in yellow precipitate is_____

59. In the chemical reaction between stoichiometric quantities of $KMnO_4$ and $MnSO_4$ in weakly alkaline solution, the number of moles of MnO_2 formed is Y. If 4.2 moles of $KMnO_4$ are consumed. The value of 2Y is…….

60. Ammonia is formed by reaction of ammonium chloride and calcium hydroxide. What is the volume of ammonia gas in litres (L) produced at 600 K and 0.82 atm pressure when 10.7 g of ammonium chloride and 10 ml of 5.0 M calcium hydroxide are mixed together for the reaction? (Given atomic mass in g mol^{-1} H, N, Cl are respectively 1, 14 and 35.5 respectively, R = 0.082 atm L mol^{-1} K^{-1})

Mathematics

PART – C

SECTION – A

(Single Choice Answer Type)

This section contains **20 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct.**

61. A function f is defined on the complex number by f(z) = (a + ib)z, where a and b are positive number. This function has the property that the image of each point in the complex plane is equidistant from that point and the origin. If |a + ib| = 8, then b^2 is

(A) $\frac{255}{4}$ (B) $\frac{245}{4}$

(C) $\frac{235}{4}$ (D) $\frac{225}{4}$

62. The smallest prime that is the fifth term of an increasing arithmetic sequence all four preceding terms also a prime is

(A) 31 (B) 29

(C) 37 (D) not possible

63. The largest two digit prime factor of $^{200}C_{100}$ is

(A) 61 (B) 59

(C) 17 (D) 60

64. The sum of all positive integers n for which $n^2 - 19n + 99$ is a perfect square is

(A) 36 (B) 37

(C) 39 (D) 38

65. Initially A, B and C had a total of 444 peanuts. C had the most peanuts and A had the least. The numbers of peanuts that each person had formed a G.P. A eats 5 of his peanuts, B eats 9 of its peanuts and C eats 25. Now the three numbers of peanuts each has forms an A.P. Find the number of peanuts A had initially

(A) 108 (B) 106

(C) 300 (D) 144

66. There is a 40% chance of rain on Saturday and a 30% chance of rain on Sunday. However, it is twice as likely to rain on Sunday if it rains on Saturday than if it does not rain on Saturday. The probability that it rains at least one day this weekend is

(A) $\frac{3}{10}$ (B) $\frac{3}{14}$

(C) $\frac{37}{70}$ (D) $\frac{33}{70}$

67. For polynomial $P(x) = 1-\frac{1}{3}x+\frac{1}{6}x^2$, define $Q(x) = P(x).P(x^3).P(x^5).P(x^7).P(x^9) = \sum_{i=0}^{50} a_i x^i$, then $\sum_{i=0}^{50}|a_i| =$

(A) $\frac{275}{37}$ (B) $\frac{243}{32}$

(C) $\frac{243}{39}$ (D) $\frac{245}{37}$

68. Let x, y and z be real numbers satisfying the system
$\log_2(xyz - 3 + \log_5 x) = 5$
$\log_3(xyz - 3 + \log_5 y) = 4$
$\log_4(xyz - 3 + \log_5 z) = 4$
then the value of $|\log_5 x| + |\log_5 y| + |\log_5 z|$ is
(A) 376 (B) 378
(C) 380 (D) 382

69. Let a and b be positive integers satisfy $\frac{ab+1}{a+b} < \frac{3}{2}$, then the maximum possible value of $\frac{a^3b^3+1}{a^3+b^3}$ is
(A) $\frac{31}{9}$ (B) $\frac{65}{9}$
(C) $\frac{65}{16}$ (D) $\frac{31}{5}$

70. In three dimensional space f(x, y, z) = xy + xz, then locus of all points which satisfy the equation f(x, y, z) = 0 is
(A) pair of perpendicular lines
(B) pair of perpendicular planes
(C) pair of a line and a plane which are perpendicular to each other
(D) pair of line and plane which are parallel to each other

71. Out of all possible 8 digit numbers formed using all the digits 0, 0, 1, 1, 2, 3, 4, 4 a number is randomly selected, probability that the selected number is odd is
(A) $\frac{5}{7}$ (B) $\frac{5}{9}$
(C) $\frac{5}{11}$ (D) $\frac{5}{14}$

72. If α, β, γ are the roots of $x^3 - 3x + 1 = 0$, then the equation whose roots are $\alpha - \frac{1}{\beta\gamma}, \beta - \frac{1}{\gamma\alpha}, \gamma - \frac{1}{\alpha\beta}$ is
(A) $x^3 - 3x + 8 = 0$ (B) $x^3 - 6x + 8 = 0$
(C) $x^3 - 9x + 8 = 0$ (D) $x^3 - 12x + 8 = 0$

73. Three Arithmetic mean, three Geometric mean and three Harmonic mean are inserted between 1 and 5. The cubic equation whose roots are 3rd A.M, 2nd G.M, and 1st H.M is
(A) $x^3 - \frac{1}{4}\left(21 + 4\sqrt{5}\right)x^2 + 5x - 5\sqrt{5} = 0$
(B) $x^3 - \frac{1}{4}\left(21 + 4\sqrt{5}\right)x^2 + \left(5 + \frac{21}{4}\sqrt{5}\right)x - 5\sqrt{5} = 0$
(C) $4x^3 - \left(21 + 4\sqrt{5}\right)x^2 + \left(15 + 21\sqrt{5}\right)x - 20\sqrt{5} = 0$
(D) $4x^3 - \left(21 + 4\sqrt{5}\right)x^2 + \left(15 + 21\sqrt{5}\right)x - 5\sqrt{5} = 0$

74. If lines $\frac{x-1}{2} = \frac{y-2}{x_1} = \frac{z-3}{x_2}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ lies in the same plane then for equation $x_1t^2 + (x_2 + 2)t + a = 0$
(A) $2x_1 - x_2 = 1$ (B) sum of roots of above equation = – 2
(C) $2x_1 + x_2 = -4$ (D) sum of root is 0

75. Let $\vec{a}$ and $\vec{b}$ are two perpendicular unit vectors. If $\vec{c}$ is another unit vector equally inclined at angle θ to the vectors $\vec{a}$ and $\vec{b}$, then the set of exhaustive values of θ is

(A) $\left(0, \frac{\pi}{2}\right)$ (B) $\left[0, \frac{\pi}{4}\right]$

(C) $\left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$ (D) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$

76. If $x + y + z = 1$ and $0 \le x, y, z \le 1$, then the maximum value of expression $27(xy + yz + zx - 2xyz)$ is equal to

(A) 7 (B) 10

(C) 8 (D) 9

77. Matrix A is given by $A = \begin{bmatrix} 6 & 11 \\ 2 & 4 \end{bmatrix}$, then the determinant of $A^{2005} - 6A^{2004}$ is

(A) 2^{2006} (B) $(-11)\, 2^{2005}$

(C) $-2^{2005} \cdot 7$ (D) $(-9)\, 2^{2004}$

78. Let a, b, c be distinct cube roots of unity and $\Delta = \begin{vmatrix} a^2+b^2 & c^2 & c^2 \\ a^2 & b^2+c^2 & a^2 \\ b^2 & b^2 & c^2+a^2 \end{vmatrix}$, then

(A) $\text{Re}(\Delta) = 0$ (B) $\text{Im}(\Delta) = 0$

(C) $\text{Re}(\Delta) + \text{Im}(\Delta) = 0$ (D) $\text{Re}(\Delta)\text{Im}(\Delta) = 4$

79. An urn contains 4 green balls and 6 blue balls. A second urn contains 16 green balls and N blue balls. A single ball is drawn at random from each urn. The probability that both balls are of the same colour is 0.58, then $(N - 140)$ is

(A) 5 (B) 4

(C) 6 (D) 7

80. A and B are two square matrix (all the entries of A and B are real such that $A^2B = BA$ and if $(AB)^{10} = A^k.B^{10}$ is true for all square matrices of same order for the value of k, then value of $k - 1020$ is

(A) 5 (B) 4

(C) 3 (D) 2

SECTION – B

(Numerical Answer Type)

This section contains **10** Numerical based questions. The answer to each question is rounded off to the nearest integer value.

81. Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$ be a polynomial whose roots are all negative integer, if $a + b + c + d = 2009$, then d is

82. If x and y are real numbers such that $\frac{(x-4)^2}{4} + \frac{y^2}{9} = 1$, then the largest possible value of $\frac{x^2}{4} + \frac{y^2}{9}$ is

83. Let p(x) be the polynomial of degree 4 with roots 1, 2, 3, 4 and leading coefficient 1. Let q(x) be the polynomial of degree 4 with roots 1, $\frac{1}{2}, \frac{1}{3}$ and $\frac{1}{4}$ and leading coefficient 1, then $\lim\limits_{x\to 1}\frac{p(x)}{q(x)}$ is equal to k, then |k| is

84. Kiran is deciding whether to visit Russia or Ukrain for the holidays. She makes her decision by rolling a regular 6-sided die. If she gets 1 or 2 she goes to Ukrain, if she rolls a 3, 4 or 5 she goes to Russia. If she rolls a 6 she rolls again, the probability that she goes to Ukrain is p, then 5p is

85. There are real numbers a, b, c and d such that -20 is a root of $x^3 + ax + b = 0$ and -21 is a root of $x^3 + cx^2 + d = 0$. These two polynomials share a complex root $m + i\sqrt{n}$, where m, n are positive integers and $i = \sqrt{-1}$, then $\frac{n}{4m}$ is

86. If $A = \begin{bmatrix} a & 2 & 3 \\ 3 & 1 & 2 \\ 2 & 1 & b \end{bmatrix}$, $B = \begin{bmatrix} 2 & -5 & d \\ -8 & -2 & 7 \\ c & 3 & -5 \end{bmatrix}$ and B = adj A and |B| =

87. If the plane containing x-axis and the point (1, 1, 1) is parallel to the line $\frac{x+1}{2k} = \frac{y-1}{k} = \frac{z}{-3}$, then the value of |k| is

88. The number of ways of arranging the letters of the word **ALGEBRA** in which either vowels or consonants, but not both appear in alphabelical order is

89. If three successive terms of a G.P. with common ratio r(r > 1) form the sides of a ΔABC and [r] denotes G.I.F., then [r] + [–r] + 5 is

90. Let a three term strictly increasing arithmetic sequence of integers are one such that the sum of the squares of the three terms equals the product of the middle term and the square of the common difference, then the sum of third term of such sequence is p, then (p – 25) is